Question Bank

Digital Electronics and Logic Design

Unit I

1. State and prove DeMorgan's first theorem using a truth table.
2. State and prove DeMorgan's second theorem using a truth table.
3. What is 1's complement?How is a negative number represented using 1's complement?
4. Represent the following using 8-bit 1's complement**:**
5. Convert the following 8-bit 1's complement numbers to decimal:
6. Convert the following 8-bit 1's complement numbers to decimal:
7. **Convert the following into canonical SOP:**

   F=A+B′C

   F=AB+C

   F=A′B+AC

   F=AB′+BC

   F=(A+B)C

8. **Convert the following into canonical POS:**

   F=A+B

   F=AB+C

   F=A′B+AC

   F=AB′+BC

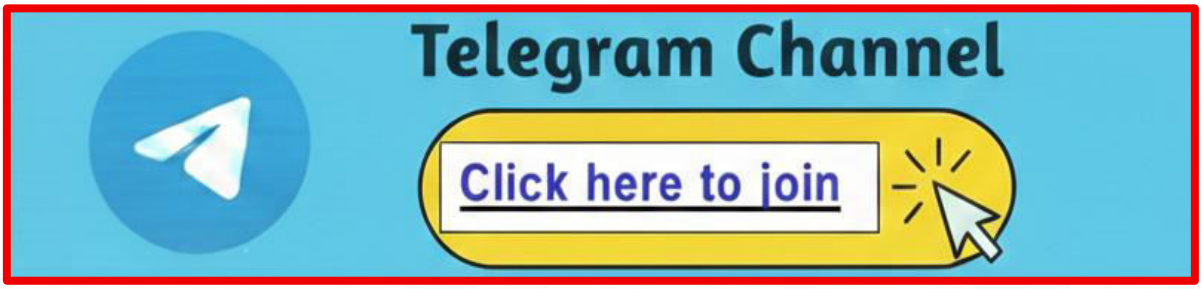
   F=A+B′C

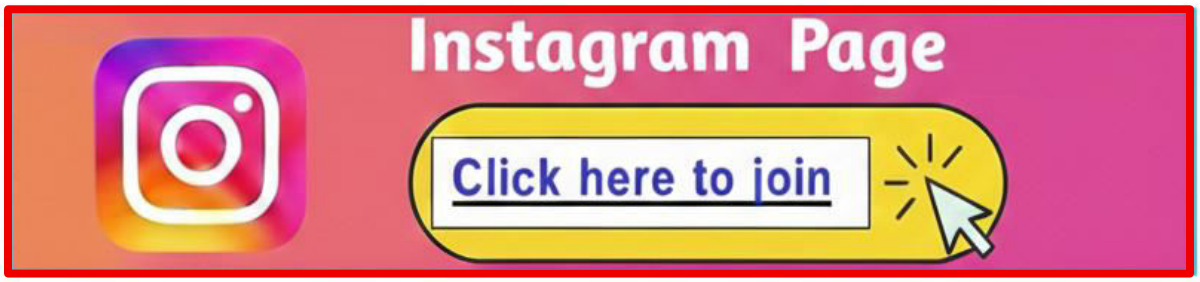
9. **Simplify using K-map:**

   a. **F(A,B)= Σm(1,3)**
   b. **F(A,B)= Σm(0,2)**
   c. **F(A,B)= Σm(1,2,3)**
   d. **F(A,B)= Σm(0,1,2)**
   e. **F(A,B)= ΠM(0,2)**
   f. **F(A,B,C)=Σm(1,3,5,7)**
   g. **F(A,B,C)=Σm(0,2,4,6)**
   h. **F(A,B,C)=Σm(1,2,3,5,7)**
   i. **F(A,B,C)=Σm(0,1,2,3)**
   j. **F(A,B,C)=Σm(2,3,6,7)**

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k. **F(A,B,C)=Σm(0,2,3,6)**
l. **F(A,B,C)=Σm(1,3,4,5,7)**
m. **F(A,B,C)=Σm(0,1,4,5,6,7)**
n. **F(A,B,C)=ΠM(0,2,4,6)**
o. **F(A,B,C)=ΠM(1,3,5,7)**
p. **F(A,B,C)=ΠM(0,1,2,3)**
q. **F(A,B,C)=ΠM(2,3,6,7)**
r. **F(A,B,C,D)=Σm(0,1,2,3)**
s. **F(A,B,C,D)=Σm(4,5,6,7)**
t. **F(A,B,C,D)=Σm(0,2,4,6,8,10,12,14)**
u. **F(A,B,C,D)=Σm(1,3,5,7,9,11,13,15)**
v. **F(A,B,C,D)=Σm(0,1,4,5,8,9,12,13)**
w. **F(A,B,C,D)=Σm(2,3,6,7,10,11,14,15)**
x. **F(A,B,C,D)=Σm(0,1,2,3,8,9,10,11)**
y. **F(A,B,C,D)=Σm(1,3,5,7,9,11,13,15)**
z. **F(A,B,C,D)=Σm(0,2,3,6,7,8,10,11,14,15)**
aa. **F(A,B,C,D)=Σm(1,2,3,5,7,9,11,13,15)**
bb. **F(A,B,C)=Σm(1,3,5,7)+d(0,2)**
cc. **F(A,B,C)=Σm(0,1,4,5)+d(2,3)**
dd. **F(A,B,C,D)=Σm(1,3,7,11,15)+d(0,2,5)**
ee. **F(A,B,C,D)=Σm(0,2,8,10,12,14)+d(1,3,9,11)**
ff. **F(A,B,C,D)=Σm(4,5,6,7,12,13,14,15)+d(0,1)**

10. State and prove both DeMorgan's theorems using truth tables.
11. Simplify using Boolean algebra:
    F=A′BC+ABC+AB′C+ABC′
12. Represent −37 in:

- 8-bit signed magnitude
- 8-bit 1's complement
- 8-bit 2's complement.

Unit II

1. **Design and Explain a** Half Adder **with truth table and logic diagram**
2. **Design and Explain a** Half Subtractor **with truth table and logic diagram.**
3. **Design and Explain a** Full Adder **with truth table and logic diagram.**

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4. **Design and Explain a** Full Subtractor **with truth table and logic diagram.**
5. **Implement a Full Adder using** two Half Adders and an OR gate**.**
6. Compare Half Adder and Full Adder.
7. Compare Half Subtractor and Full Subtractor.
8. **Explain the working of a** 4-bit binary parallel adder**.**
9. **Explain the operation and pin configuration/function of** IC 7483**.**
10. **Explain the** Carry Look-Ahead Adder**. Derive the carry equations using Generate and Propagate concepts**.
11. **Design a 4-bit Binary-to-Gray code converter** using logic gates.
12. **Design a 4-bit Gray-to-Binary code converter** using logic gates.
13. **Explain 4:1 Multiplexer with truth table, logic circuit and Boolean expression**
14. **Explain 1:4 Demultiplexer with truth table and logic circuit.**
15. Implement a given Boolean function using a multiplexer.
16. **Design and explain a 2-bit magnitude comparator.**
17. Compare MUX, DEMUX based on their functions and applications.
18. **Prepare all types of Multiplexers and De- multiplexers**

Unit III

1. **Explain the working of** SR Flip-Flop **with circuit diagram, truth table**
2. **Explain the working of clocked** SR Flip-Flop **with circuit diagram, truth table**
3. **Explain the working of** JK Flip-Flop **with circuit diagram, truth table**
4. **Explain the working of** D Flip-Flop **with circuit diagram, truth table**
5. **Explain the working of** T Flip-Flop **with circuit diagram, truth table and characteristic equation**
6. **Explain the** Master-Slave JK Flip-Flop **with a neat circuit diagram**
7. **What are** Preset and Clear inputs**? Explain their operation in a flip-flop with suitable truth table.**
8. Explain the **truth table and excitation table of D Flip-Flop**.
9. Construct the **excitation table of JK Flip-Flop from its truth table** and explain each condition.
10. Explain the procedure for **conversion of SR Flip-Flop into JK Flip-Flop** using excitation tables and logic gates
11. Explain the **conversion of D Flip-Flop into T Flip-Flop** with suitable logic diagram and excitation table
12. Explain the general **step-by-step procedure for converting one type of flip-flop into another type**

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13. **Explain the working of a** Serial-In Serial-Out (SISO) shift register **with block diagram**
14. **Explain the working of a** Serial-In Parallel-Out (SIPO) shift register **with suitable diagram**
15. **Explain the working of a** Parallel-In Parallel-Out (PIPO) register **with block diagram**
16. **Explain the working of a** Parallel-In Serial-Out (PISO) shift register **with suitable diagram**
17. Explain the applications of **shift registers** in digital systems
18. **Explain the working of a** Ring Counter **with a neat circuit diagram**
19. Explain the **advantages, limitations and applications of Ring Counters**
20. **Explain the working principle of a** Johnson Counter **with a suitable circuit diagram**
21. **Explain the working of a** BCD Counter **with suitable logic diagram and state sequence.**
22. Explain the applications of **BCD Counters** in digital electronic systems
23. **Design a** MOD-10 BCD Counter **using JK Flip-Flops and explain the counting sequence**.
24. Comparison of **Ring Counter and Johnson Counter**.
25. What is Race around condition.

Unit IV

1. Explain the basic structure of an ASM chart with suitable diagrams and examples.
2. Describe the construction of an ASM chart step by step, explaining all symbols and notations.
3. Compare Moore and Mealy machines.
4. . Compare FSM and ASM.
5. Differentiate between state boxes and decision boxes in an ASM chart.
6. Compare unipolar and bipolar logic families.
7. Compare TTL and CMOS logic families.
8. Draw an ASM chart for a sequential circuit that controls a simple two-state process.
9. Develop a state table from a given ASM chart.
10. Explain the construction and operation of a CMOS inverter with a neat circuit diagram.
11. Explain the advantages, disadvantages, and applications of TTL and CMOS logic families.

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12. Explain the classification of logic families into unipolar and bipolar families with examples.
13. Discuss the various characteristics of digital ICs,
14. Compare different logic families based on speed, power consumption, noise immunity, and fan-out.

UNIT V

1. Explain the architecture of PAL with a neat diagram.
2. Differentiate between PLA and PAL.
3. Explain the role of the programmable AND array and fixed OR array in PAL.
4. Describe how Boolean expressions are implemented using PAL.
5. Explain the advantages and limitations of PAL.
6. Discuss the applications of PAL in digital circuit design.
7. Explain the programming procedure of PAL step by step with a suitable example.
8. Explain the design of combinational logic circuits using PAL with a neat diagram.
9. Describe the conversion of a truth table into Boolean expressions for PLD implementation.
10. Describe the implementation of multiple-output functions using PLA.
11. Explain the implementation of a code converter using a PLD.
12. Describe the steps for implementing a decoder using a PLD.
13. Explain the complete procedure for designing a combinational circuit using a PLD, from truth table to final implementation.
14. Design a combinational circuit using PLA for the following functions:

    $F1(A,B,C) = \Sigma m\ (1,2,5,7)$

    $F2(A,B,C) = \Sigma m\ (0,2,6,7)$
15. Design a 2-to-4 decoder using a PLD. Write the truth table, Boolean equations, and implementation steps.
16. Design a binary-to-Gray code converter using a PLD. Show the truth table and minimized Boolean equations.
17. Explain the design and implementation of a full adder using PLA or PAL.
18. Design a combinational circuit that produces the BCD-to-Excess-3 code conversion using a PLD.

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